How Far Does Treatment Go?

A Global Analysis of ART Treatment Coverage and HIV Mortality Patterns

STA198 Team: Natalie Alba, Ariella Maroni, Jessica Perez, Ali Wilson

April 22, 2026

```r
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE)
options(digits = 3)
```

Introduction

HIV/AIDS remains one of the most significant global public health challenges, affecting millions of individuals worldwide and continuing to place a substantial burden on healthcare systems. According to the World Health Organization, approximately 39 million people are currently living with HIV globally, and HIV-related illness continue to account for hundreds of thousands of deaths each year (World Health Organization, 2023). Despite major advances in treatment and prevention, the burden of HIV/AIDS is not evenly distributed across countries, with some regions experiencing substantially higher mortality rates than others.

HIV/AIDS outcomes at a global scale are closely linked to access to antiretroviral therapy (ART) and other structural factors that vary across countries. ART suppresses the replication of the virus, allowing individuals living with HIV to maintain healthier immune systems and significantly lowering the risk of progression to AIDS and death (World Health Organization, 2023). As a result, increasing access to ART has become a central focus of global health initiatives, but ART varies widely across countries due to differences in healthcare infrastructure, economic resources, and public health policy.

Because of this variation, it is important to understand whether higher ART coverage is actually associated with lower HIV/AIDS death counts when comparing countries. In addition, countries with larger populations of people living with HIV are naturally expected to have higher numbers of deaths, regardless of treatment coverage. This makes it necessary to account for population size when evaluating the relationship between ART coverage and mortality. HIV/AIDS death counts are also highly skewed across countries, with a small number of countries contributing disproportionately large values. These characteristics demonstrate the need for both adjusted models and transformations.

In this study, we examine the association between ART coverage and HIV/AIDS deaths across countries using country-level data. We explore both unadjusted and adjusted relationships, incorporating the number of individuals living with HIV as a control variable and applying log transformations to better capture the relationship in the data. This leads to our primary research question:

Is higher antiretroviral therapy (ART) coverage associated with lower HIV/AIDS death counts across countries?

Data

The dataset used in this analysis was constructed by merging three country-level datasets containing HIV/AIDS indicators: HIV/AIDS deaths, the number of people living with HIV, and antiretroviral therapy (ART) coverage among people living with HIV. The original data were accessed from a GitHub repository (https://github.com/srivi15/HIV-AIDS---Data-Analysis) and are derived from publicly available sources, including the World Health Organization (WHO), UNESCO, and Kaggle.

Prior to merging, each dataset was cleaned and standardized to ensure consistency across variables. Variable names were reformatted, and the data were aggregated at the country level. For both the HIV/AIDS deaths and HIV population datasets, median estimates were used to represent each country. The ART dataset was also cleaned to only the relevant variable measuring the percentage of individuals living with HIV who are receiving treatment.

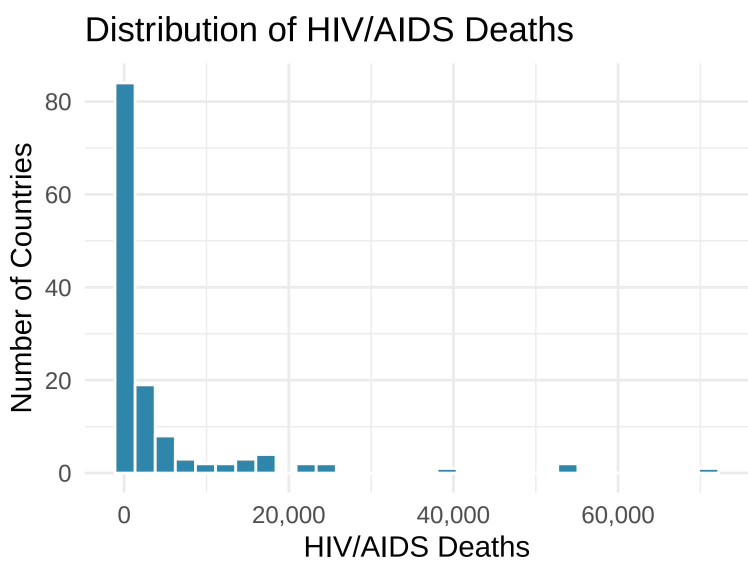
The final dataset contains 170 countries, with each observation representing a single country. The primary variables used in the analysis are HIV/AIDS deaths, number of people living with HIV, and ART coverage (percentage of people living with HIV receiving treatment). All variables are continuous numerical estimates.

Table 1: Missing values by variable.

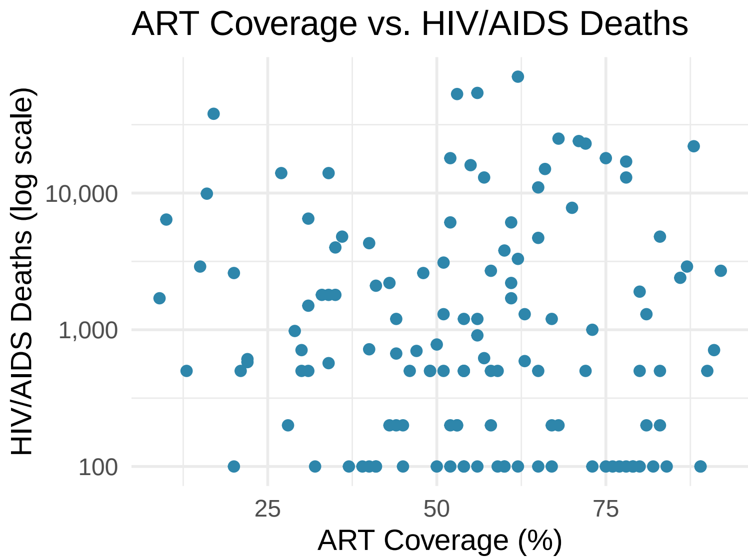
Variable	Missing..n.	Missing....
HIV/AIDS Deaths	37	21.8%
HIV Population	32	18.8%
ART Coverage	34	20.0%

Missing values are present in all main variables: HIV/AIDS deaths (37 missing, ~21.8%), HIV population (32 missing, ~18.8%), and ART coverage (34 missing, ~20%). These missing values likely reflect the real-world issue of incomplete reporting in global health surveillance systems. For the analysis, missing data were handled using listwise deletion, meaning observations with missing values were automatically excluded when fitting each model.

The following figures illustrate the distribution of HIV/AIDS deaths and the unadjusted relationship between ART coverage and mortality across countries.



Distribution of HIV/AIDS deaths across countries. The strong right skew, with a small number of countries contributing very high death counts, motivates the use of a log transformation in the regression models.



ART coverage (%) versus HIV/AIDS deaths across countries on a log scale. The pattern suggests a potential negative association that motivates the controlled regression analysis.

Methods

To examine the relationship between ART coverage and HIV/AIDS deaths across countries, we conducted a series of statistical analyses. We begin with a correlation test to assess the unadjusted relationship between ART coverage and deaths. We then fit a sequence of regression models, including a simple linear regression, a multiple linear regression controlling for the number of people living with HIV, and a log-transformed model to account for skewness in the data. Each model builds on limitations of the previous one and is interpreted in the sections below.

Linear regression was chosen as the primary modeling framework because the outcome of interest (HIV/AIDS deaths) is a continuous variable at the country level, and the goal is to estimate the association between ART coverage and mortality while controlling for HIV population size. Although HIV/AIDS deaths are counts and could be modeled using Poisson or negative binomial regression, the country-level aspect and the severe overdispersion in the data make ordinary least squares with a log transformation a better alternative.

The correlation analysis provides an initial measure of the strength and direction of the linear association between ART coverage and HIV/AIDS deaths, but it does not account for differences across countries. To address this, a simple linear regression model was first fit with HIV/AIDS deaths as the response variable and ART coverage as the predictor. This serves as a baseline model for the relationship without any adjustments.

A multiple linear regression model was then used to include both ART coverage and the number of individuals living with HIV, allowing us to control for differences in HIV burden across countries. This is important because countries with larger HIV-positive populations are expected to have higher numbers of deaths regardless of ART coverage.

The distribution of HIV/AIDS deaths is strongly right-skewed, with a small number of countries accounting for disproportionately large values, which motivated the decision for the log transformation used in Model 3. The log-log specification linearizes the relationship between HIV population size and deaths, and allows the ART coverage coefficient to be interpreted as a percentage change in deaths per unit increase in coverage.

Model assumptions for the final model were assessed using a residuals versus fitted values plot to evaluate linearity and constant variance, and a histogram of residuals to evaluate normality. These diagnostics were applied to Model 3, as it is the primary inferential model.

Results

Correlational Analysis

The correlation between ART coverage and HIV/AIDS deaths was very weak ($r = 0.023$), indicating little to no linear relationship when examined alone. However, this result may be misleading because it does not account for differences in the number of people living with HIV across countries, which can strongly influence death counts. This suggests that a controlled regression analysis is necessary.

Simple Linear Regression

Table 2: Simple OLS. R-squared = 5e-04 | N = 131

Term	Estimate	Std. Error	p-value
Intercept	3960.2	2697.7	0.145
ART Coverage (%)	12.1	46.2	0.794

A simple linear regression was conducted to examine the relationship between ART coverage and HIV/AIDS deaths. The results showed no significant relationship between ART coverage and deaths (beta = 12.07, $p = 0.794$). Additionally, the model explained virtually none of the variation in deaths (R^2 = 0.0005), indicating that ART coverage alone is not a meaningful predictor of HIV/AIDS mortality. This suggests that other factors, such as the size of the HIV-positive population, may play a more important role.

Multiple Linear Regression

Table 3: Controlled OLS. R-squared = 0.753 | N = 131

Term	Estimate	Std. Error	p-value
Intercept	4020.974	1347.080	0.003
ART Coverage (%)	-40.546	23.232	0.083
HIV Population	0.012	0.001	<0.001

A multiple linear regression model was used to examine the relationship between ART coverage and HIV/AIDS deaths while controlling for the number of people living with HIV. The results showed that ART coverage had a negative association with deaths (beta = -40.55, $p = 0.083$), suggesting a potential negative relationship between ART coverage and HIV/AIDS deaths. Although this relationship was not statistically significant at the 0.05 level, it indicates a possible downward trend.

In contrast, the number of people living with HIV was a highly significant predictor of deaths (beta = 0.01208, $p < 2e-16$), indicating that countries with larger HIV-positive populations experience substantially more deaths.

The model demonstrated strong explanatory power (R^2 = 0.753), meaning that approximately 75% of the variation in HIV/AIDS deaths across countries is explained by these variables.

All in all, while ART coverage alone did not appear to influence deaths in the simple model, the controlled regression suggests that increased ART coverage is associated with reduced HIV/AIDS mortality when accounting for differences in HIV population size. This is consistent with the hypothesis that higher ART coverage is associated with lower HIV-related deaths globally.

Log-Transformed Model

Table 4: Log-Transformed Model. R-squared = 0.9 | N = 131

Term	Estimate	Std. Error	p-value
Intercept	-0.663	0.275	0.017
ART Coverage (%)	-0.014	0.003	<0.001
log(HIV Population)	0.800	0.024	<0.001

A log-transformed multiple regression model was used to better account for skewness in the data and examine the relationship between ART coverage and HIV/AIDS deaths. The model takes the form:

$$\log(\text{deaths}) = \beta_0 + \beta_1(\text{art\_coverage}) + \beta_2 \log(\text{hiv\_population}) + \epsilon$$

The results showed that ART coverage had a statistically significant negative association with log(HIV/AIDS deaths) ($\beta = -0.0143$, p = 9.96e-08), after controlling for log(HIV population size). This means that for each 1 percentage point increase in ART coverage, a country's expected HIV/AIDS deaths are multiplied by a factor of e^(-0.0143) 0.986. This is about a 1.4% decrease in deaths per additional percentage point of ART coverage.

log(HIV population size) was also highly significant ($\beta = 0.800$, $p < 2\text{e-}16$). This confirms that population size remains the main contributor to death counts.

This model explained 90.0% of the variation in log(HIV/AIDS deaths) ($R^2 = 0.900$), a substantial improvement over both the simple ($R^2 = 0.0005$) and controlled ($R^2 = 0.753$) untransformed models, suggesting the log transformation better captures the underlying relationships in the data.

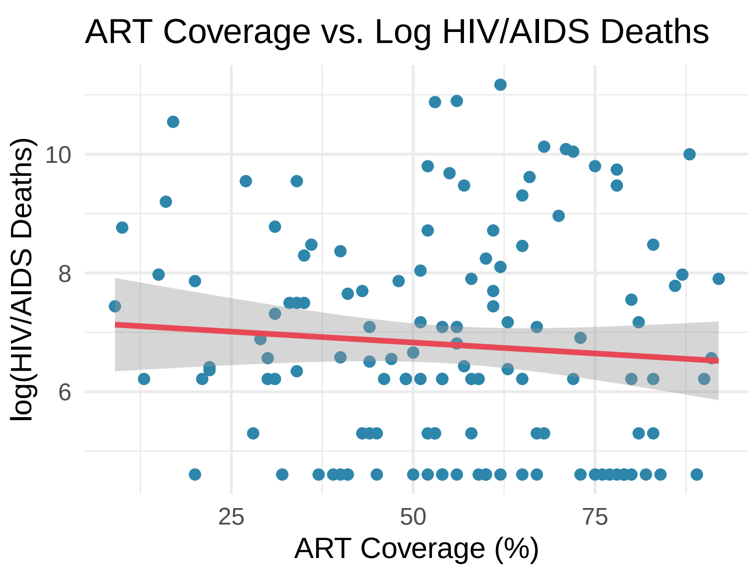


Figure 1: ART coverage (%) versus log(HIV/AIDS deaths) across countries, with fitted regression line and 95% confidence band from Model 3. The negative slope indicates that higher ART coverage is associated with lower mortality after controlling for HIV population size.

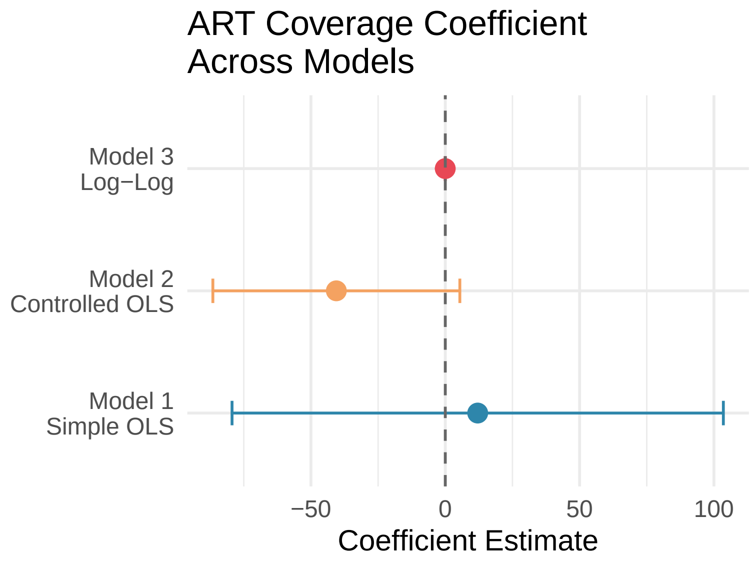


Figure 2: ART coverage coefficient estimates with 95% confidence intervals across all three model specifications. The interval crosses zero in Models 1 and 2, indicating non-significance. Only in Model 3 does the estimate become statistically significant.

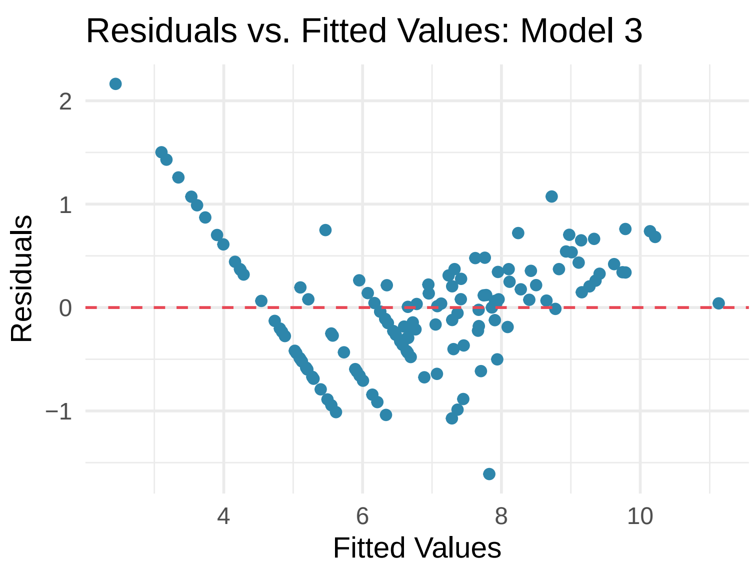


Figure 3: Residuals versus fitted values for Model 3. Points are scattered randomly around zero with no systematic pattern, supporting the linearity and constant variance assumptions.

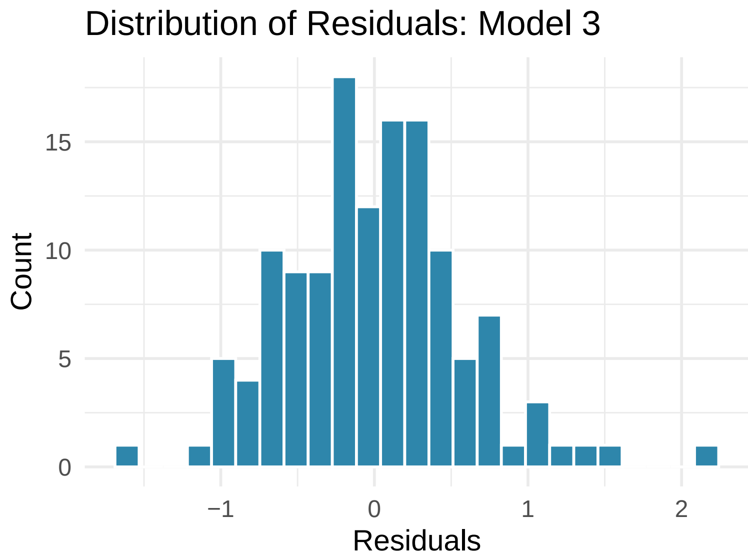


Figure 4: Distribution of residuals from Model 3. The approximately symmetric, bell-shaped distribution centered near zero supports the normality assumption.

Discussion

This analysis examined the relationship between ART coverage and HIV/AIDS deaths across countries using multiple statistical models. The results indicate that the relationship between ART coverage and mortality depends strongly on model specification. In the initial correlation and simple regression models, ART coverage showed little to no association with HIV/AIDS deaths. However, these results may be misleading because they do not account for differences in the size of the HIV-positive population across countries.

After controlling for the number of people living with HIV, ART coverage showed a negative association with HIV/AIDS deaths, suggesting that higher treatment coverage is associated with fewer deaths. Although this relationship was not statistically significant at the 0.05 level in the multiple linear regression model, the direction and magnitude of the effect suggest a potential downward trend. The log-transformed model provided the strongest evidence of a relationship, showing a statistically significant negative association between ART coverage and HIV/AIDS deaths after adjusting for HIV population size.

These results suggest that model specification plays an important role in identifying the relationship between ART coverage and HIV/AIDS outcomes. In particular, the improved fit of the log-transformed model indicates that accounting for skewness in both deaths and HIV population provides a more appropriate representation of the data structure. Alternative approaches such as Poisson or negative binomial regression could also be explored in future work, as these may be more appropriate for count outcomes.

Despite these findings, several limitations should be considered. The analysis is based on observational, country-level data, which limits the ability to make causal conclusions about the relationship between ART coverage and HIV/AIDS deaths. The presence of missing data across key variables may introduce bias and affect the reliability of estimates. The models also do not account for other potentially important factors, such as healthcare infrastructure, economic conditions, or access to prevention and testing services, which may influence HIV/AIDS outcomes across countries. Future work could improve this analysis by incorporating additional country-level variables and using longitudinal data to examine how changes in ART coverage are associated with changes in mortality over time. These extensions would provide a more complete understanding of the drivers of HIV/AIDS outcomes.

Conclusion

In conclusion, this study examined the relationship between ART coverage and HIV/AIDS deaths across countries using correlation analysis and multiple regression models. While initial analyses suggested little to no association between ART coverage and mortality, more refined models that accounted for HIV population size and data skewness revealed a significant negative relationship. In particular, the log-transformed model provided the strongest evidence that higher ART coverage is associated with lower HIV/AIDS deaths. These findings demonstrate the importance of expanding access to antiretroviral therapy as a key strategy for reducing HIV/AIDS mortality at a global scale.

References

Srividhya R. (n.d.). HIV AIDS – Data Analysis and Visualization. GitHub. https://github.com/srivi15/HIV-AIDS---Data-Analysis

World Health Organization. (2023). HIV/AIDS. https://www.who.int/news-room/fact-sheets/detail/hiv-aids